

- 17 A tailor shop makes dresses and shirts.
The following table shows the different components of manufacturing each clothing item.

	Labour (Hours)	Cloth (Metres)	Buttons (Rows)
Dress	7	5	2
Shirt	4	5	3

Labour costs \$8 per hour, cloth costs \$11 per metre and buttons cost \$3 per row.

It is given that $A = \begin{pmatrix} 7 & 5 & 2 \\ 4 & 5 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 8 \\ 11 \\ 3 \end{pmatrix}$ and $C = AB$.

- (a) (i) Evaluate C .

Answer [1]

- (ii) Explain what your answer in (a)(i) represents.

Answer

.....

.....

.....

[1]

(b) In addition, $D = \begin{pmatrix} 100 & 150 \end{pmatrix}$.

(i) Evaluate DC .

Answer [1]

(ii) Explain what your answer in (b)(i) represents.

Answer

.....

.....

.....

[1]

18 Factorise each of the following completely.

(a) $(2a)^2 - 4ab + 28a + (b-7)^2$.

Answer [2]

(b) $xy^2 - 9x + 3y - y^2$.

Answer [3]